Density Functional Theory: Transition States, Reaction Mechanisms, and Energy Profiles

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Recap from Lecture 1

What we covered:

- Molecular Hamiltonian & Born–Oppenheimer approximation
- HF theory vs. DFT
- Kohn–Sham DFT and E_{XC}
- Jacob's ladder of functionals
- Potential energy surface (PES)
- Harmonic approximation & normal modes
- Partition functions from DFT
- Thermodynamic properties (ΔG, ΔH, ΔS)

Key result:

$$G = E_{\text{elec}} + E_{\text{ZPE}} + H_{\text{thermal}} - TS$$

Today:

How do we use DFT to study **chemical reactions**?

(Recall Week 1: thermo is empirical. DFT makes it predictive.)

- Finding transition states
- Building reaction energy profiles
- Computing rate constants
- Understanding selectivity

Lecture 2 Overview

Part I: Transition State Theory

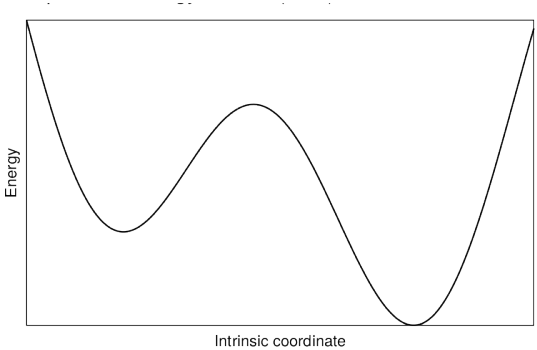
1. Transition states on the PES
2. Saddle points & imaginary frequencies
3. Transition state theory (TST)
4. The Eyring equation
5. Thermodynamic activation parameters

Part II: Computational Practice

6. Finding transition states computationally
7. Intrinsic reaction coordinate (IRC)
8. Building reaction energy profiles
9. Selectivity and competing pathways
10. Case studies in organic & inorganic chemistry

Goal: Learn how to use DFT to compute reaction barriers, rate constants, and predict which products form and why.

Chemical Reactions on the PES



A chemical reaction is a path on the PES from **reactant minimum** to **product minimum**.

Key features:

- **Minima**: stable species (reactants, products, intermediates)
- **Saddle points**: **transition states** (TS) – highest point along the minimum energy path
- **Reaction coordinate**: the path connecting R $\rightarrow$ TS $\rightarrow$ P

The **activation energy** E_a is the energy difference between the TS and reactants.

Recall the curse of dimensionality from Week 3: $\sim 10^{3N-6}$ grid points on the PES. Finding a TS on this surface is the challenge we tackle today.

What is a Transition State?

Definition

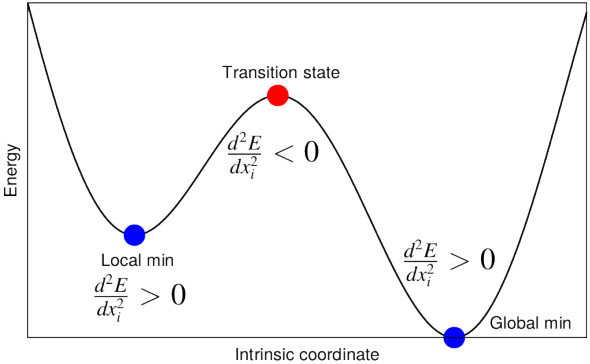
A **transition state** (TS) is a first-order saddle point on the PES:
– a **maximum** along the reaction coordinate
– a **minimum** in all other directions

Mathematical characterization:

- $\nabla E = \mathbf{0}$ (stationary point, like a minimum)
- Hessian has **exactly one negative eigenvalue**
- $\Rightarrow$ exactly **one imaginary frequency**
- The eigenvector of the imaginary frequency shows the **reaction coordinate** – the motion that converts reactants to products

	Minimum	**Transition State**
Gradient	$\mathbf{0}$	$\mathbf{0}$
Hessian eigenvalues	All positive	One negative
Imaginary frequencies	0	**1**

Imaginary Frequencies – The Signature of a TS



At a minimum:

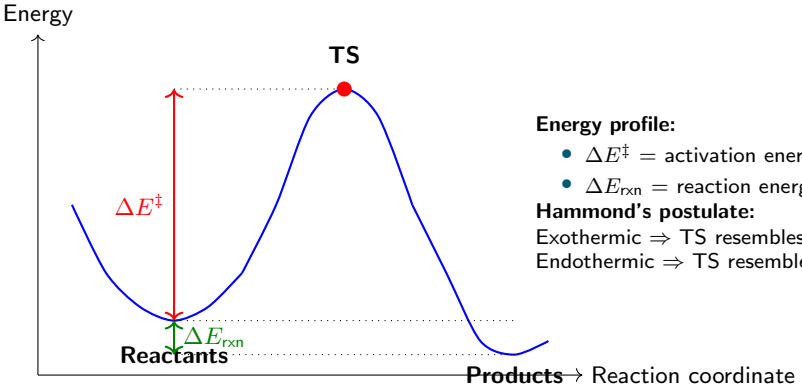
- All frequencies are **real** (positive)
- All Hessian eigenvalues > 0
- The structure is a **stable** species

At a transition state:

- One frequency is **imaginary** ($i\nu$)
- One Hessian eigenvalue < 0
- The mode shows **bond breaking/forming**

Always verify your TS: visualize the imaginary mode – does it correspond to the reaction you expect?

The Reaction Coordinate



Energy profile:

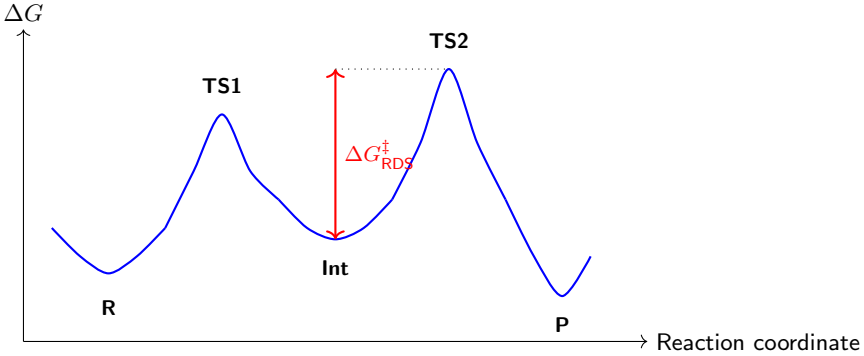
- $\Delta E^{\ddagger}$ = activation energy
- ΔE_{rxn} = reaction energy

Hammond's postulate:

Exothermic $\Rightarrow$ TS resembles reactants
Endothermic $\Rightarrow$ TS resembles products

Multi-Step Reactions

Many reactions proceed through **intermediates**:



- The **rate-determining step (RDS)** has the **highest barrier** relative to the lowest preceding intermediate
- All minima and TS must be optimized and characterized by frequency calculations
- The overall barrier is from the **resting state** to the **highest TS**

Transition State Theory (TST) – Assumptions

Eyring's TST (1935) connects the microscopic PES to macroscopic rate constants:

Key assumptions:

1. The Born–Oppenheimer approximation is valid
2. Classical nuclear motion along the reaction coordinate
3. **Quasi-equilibrium:** Reactants and the TS are in quasi-thermodynamic equilibrium:

$$\mathrm{A} + \mathrm{B} \rightleftharpoons [\mathrm{AB}]^{\ddagger} \rightarrow \text{Products}$$

4. The TS is crossed only once (no re-crossing)
5. Separability of the reaction coordinate from other degrees of freedom

TST provides an upper bound to the true rate constant (re-crossing lowers the rate).

The Eyring Equation

The Eyring Equation

$$\boxed{k(T) = \kappa \frac{k_B T}{h} \mathrm{e}^{-\Delta G^{\ddagger}/RT}}$$

- $k(T)$ = rate constant at temperature T
- κ = transmission coefficient (≤ 1, often assumed $= 1$)
- $k_B T/h$ = frequency factor ($\approx 6.2 \times 10^{12}$ s^{-1} at 298 K)
- $\Delta G^{\ddagger}$ = Gibbs free energy of activation

Alternatively, in terms of $\Delta H^{\ddagger}$ and $\Delta S^{\ddagger}$:

$$k(T) = \kappa \frac{k_B T}{h} \mathrm{e}^{\Delta S^{\ddagger}/R} \mathrm{e}^{-\Delta H^{\ddagger}/RT}$$

Compare with Arrhenius: $k = A\mathrm{e}^{-E_a/RT} \quad \Rightarrow \quad E_a \approx \Delta H^{\ddagger} + RT$

The $\Delta G^{\ddagger}$ here is computed from the same $q_{trans} \cdot q_{rot} \cdot q_{vib} \cdot q_{elec}$ partition function machinery you've been building all term — now applied to the TS.

Activation Parameters from DFT

From DFT frequency calculations on reactants and TS:

$$\Delta E^{\ddagger} = E_{\text{elec}}(\text{TS}) - E_{\text{elec}}(\text{R})$$ electronic barrier

$$\Delta E_0^{\ddagger} = \Delta E^{\ddagger} + \Delta E_{\text{ZPE}}^{\ddagger}$$ ZPE-corrected barrier

$$\Delta H^{\ddagger} = \Delta E_0^{\ddagger} + \Delta H_{\text{thermal}}^{\ddagger}$$ enthalpy of activation

$$\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$$ free energy of activation

Important: The TS has $3M - 7$ real vibrational modes (one mode is the reaction coordinate and is **excluded** from $q_{\text{vib}}^{\ddagger}$). This is already handled automatically by the frequency calculation!

Deriving the Eyring Equation

Starting from TST quasi-equilibrium:

$$K^{\ddagger} = \frac{[\mathrm{TS}]}{[\mathrm{R}]} = \frac{q^{\ddagger}}{q_{\mathrm{R}}} \mathrm{e}^{-\Delta E_0^{\ddagger}/k_B T}$$

The TS partition function is factored:

$$q^{\ddagger} = \underbrace{q^{\ddagger}_{\text{react. coord.}}}_{\text{1D translation}} \times q^{\ddagger}_{\text{remaining}}$$

The reaction coordinate contribution (particle in a box of width δ):

$$q^{\ddagger}_{\text{react. coord.}} = \frac{\sqrt{2\pi\mu k_B T}}{h}\,\delta$$

The rate of crossing the barrier:

$$k = \frac{k_B T}{h} \cdot \frac{q^{\ddagger}_{\text{remaining}}}{q_{\mathrm{R}}} \cdot \mathrm{e}^{-\Delta E_0^{\ddagger}/k_B T} = \frac{k_B T}{h} K^{\ddagger}_{\text{thermo}} = \frac{k_B T}{h} \mathrm{e}^{-\Delta G^{\ddagger}/RT}$$

Numerical Example: How Barrier Height Affects Rate

Using the Eyring equation at $T = 298$ K:

$$k = \frac{k_B T}{h} \mathrm{e}^{-\Delta G^{\ddagger}/RT} \approx 6.2 \times 10^{12} \cdot \mathrm{e}^{-\Delta G^{\ddagger}/RT} \ \ \mathrm{s}^{-1}$$

$\Delta G^{\ddagger}$ (kcal/mol)	k (s^{-1})	**Half-life**
5	1.4×10^{9}	∼0.5 ns
10	3.1×10^{5}	∼2 μs
15	6.8×10^{1}	∼10 ms
20	1.5×10^{-2}	∼46 s
25	3.3×10^{-6}	∼2.4 days
30	7.3×10^{-10}	∼30 years
35	1.6×10^{-13}	∼140,000 years

1 kcal/mol error in $\Delta G^{\ddagger}$ changes the rate by a factor of ∼5.5!
This is why accurate barriers are so important (and challenging).

Reaction Path Degeneracy

If there are σ_{rxn} equivalent pathways, the rate is multiplied:

$$k_{\text{obs}} = \sigma_{\text{rxn}} \cdot k_{\text{single path}}$$

Examples:

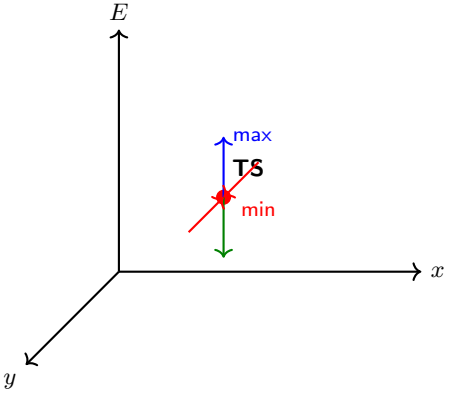
- **H-abstraction from CH_4:** 4 equivalent C–H bonds $\Rightarrow \sigma_{\text{rxn}} = 4$
- **Deprotonation of C_2H_6:** 6 equivalent C–H bonds $\Rightarrow \sigma_{\text{rxn}} = 6$
- **S_N2 at CH_3Cl:** only 1 backside attack $\Rightarrow \sigma_{\text{rxn}} = 1$

$\sigma_{\text{rxn}} = \dfrac{\sigma_{\text{R}}}{\sigma_{\text{TS}}}$, where σ are the rotational symmetry numbers.
This accounts for the fact that symmetry-equivalent paths are already included in the rotational partition functions.

TS Optimization – The Challenge

Finding a TS is **much harder** than finding a minimum:

- A minimum is a "bowl" – optimization naturally rolls downhill
- A TS is a "saddle" – must go **uphill** along reaction coordinate and **downhill** in all others
- No guarantee of finding the **right** TS
- Multiple TS may exist for the same reaction
- Requires a **good initial guess**



Saddle point: maximum in one direction, minimum in all others.

Method 1: Relaxed PES Scan

Approach: Systematically vary the reaction coordinate while optimizing all other coordinates.

Procedure:

1. Choose a coordinate that changes during the reaction (e.g., a bond length)
2. Fix this coordinate at a series of values between reactant and product
3. At each point, optimize all other coordinates ("relaxed scan")
4. The maximum of the scan gives an **approximate TS geometry**
5. Use this as the starting point for a proper TS optimization

Advantages:

- Gives a visual picture of the reaction path
- Helps identify the right reaction coordinate
- Good for generating TS guesses

Limitation: The scan coordinate may not be the true reaction coordinate.

Method 2: QST2 and QST3

Quadratic Synchronous Transit – uses knowledge of reactant and product geometries:

QST2: Requires reactant + product geometries

- Interpolates a path and finds the maximum along it
- Then refines using eigenvector-following

QST3: Requires reactant + TS guess + product geometries

- Better convergence with a reasonable TS guess
- The TS guess can come from a relaxed scan

> **Critical:** Atom ordering must be **consistent** between reactant, TS, and product structures! The algorithm maps atoms by index.

In Gaussian: `opt=(QST2)` or `opt=(QST3)`
In ORCA: `NEB-TS` (nudged elastic band → TS)

Method 3: Nudged Elastic Band (NEB)

NEB finds the **minimum energy path (MEP)** between two endpoints:

1. Generate a "chain" of images (structures) between reactant and product
2. Connect images with "springs" (elastic band)
3. Optimize the band: minimize perpendicular forces, maintain spacing
4. The highest-energy image approximates the TS

Climbing-Image NEB (CI-NEB):

- The highest image is driven **uphill** along the path
- Converges to the true saddle point

Advantages:

- Does not require a TS guess
- Finds the MEP, not just the TS
- Good for complex reactions where the coordinate is unclear

Disadvantage: Computationally expensive (many images, many optimizations)

Method 4: Eigenvector Following (Berny Algorithm)

Direct TS optimization: Follow the lowest eigenvector of the Hessian uphill.

Procedure:

1. Start with a TS guess (from scan, NEB, or chemical intuition)
2. Compute or estimate the Hessian
3. Maximize along the eigenvector with the **most negative** eigenvalue
4. Minimize along all other eigenvectors
5. Iterate until convergence

In Gaussian: `opt=(TS,CalcFC)` or `opt=(TS,CalcAll)`

- `CalcFC`: compute the Hessian at the first step
- `CalcAll`: compute the Hessian at every step (expensive but reliable)

Warning: The initial guess **must** already be close to the TS! The algorithm follows the **nearest** saddle point, which may not be the one you want.

Verifying the Transition State

After finding a TS, you MUST verify it:

1. **Frequency calculation:** Confirm exactly **one imaginary frequency**
   - Zero imaginary → it's a minimum, not a TS!
   - Two or more → higher-order saddle point, not a TS
2. **Visualize the imaginary mode:** Does it correspond to the expected bond breaking/forming?
   - For S_N2: nucleophile approaching + leaving group departing
   - For H-transfer: H moving between donor and acceptor
3. **IRC calculation:** Follow the imaginary mode in both directions
   - Forward → should reach products
   - Backward → should reach reactants
   - This **proves** the TS connects the right endpoints

Skipping verification is the #1 source of errors in computational reaction studies!

Intrinsic Reaction Coordinate (IRC)

The IRC is the steepest-descent path from the TS in mass-weighted coordinates.

$$\frac{d\mathbf{q}}{ds} = -\frac{\nabla V(\mathbf{q})}{|\nabla V(\mathbf{q})|}$$

where s is the arc length along the path and $\mathbf{q}$ are mass-weighted coordinates.

What IRC gives you:

- Confirms the TS connects the **correct** reactant and product
- Provides the minimum energy path (MEP)
- Energy profile along the reaction coordinate
- Structural changes along the reaction path

In Gaussian: `IRC=(CalcFC,MaxPoints=50)`
In ORCA: `%irc MaxIter 50 end`

Tip: Always run IRC in **both directions** from the TS (forward and reverse).

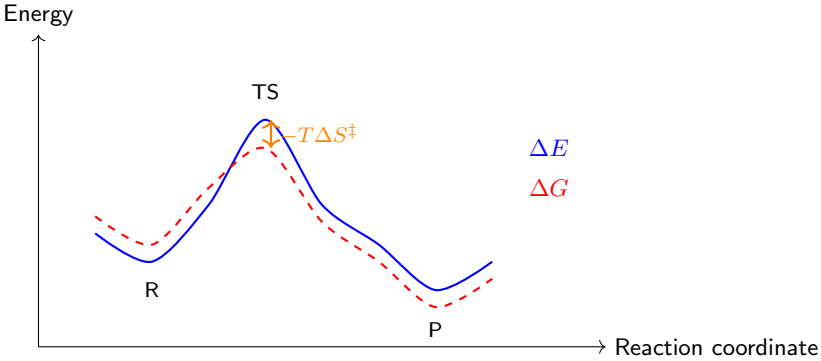
Building a Complete Reaction Energy Profile

Step-by-step workflow:

1. **Optimize reactants** – frequency calculation confirms minimum (0 imaginary freq.)
2. **Optimize products** – frequency calculation confirms minimum
3. **Find and optimize TS** – frequency calculation confirms saddle point (1 imaginary freq.)
4. **Run IRC** – confirm TS connects reactants to products
5. **Optimize intermediates** (if multi-step) – frequency calculation confirms each
6. **Compute** $\Delta G^{\ddagger}$ and ΔG_{rxn} from the thermochemistry output

All species must be computed at the same level of theory (same functional, same basis set, same temperature/pressure) for meaningful energy differences!

Electronic Energy vs. Free Energy Profiles



Why $\Delta E \neq \Delta G$:

- ZPE: usually lowers barrier by 1–3 kcal/mol
- Thermal: depends on T and frequencies
- Entropy: TS tighter than reactants $\Rightarrow \Delta S^{\ddagger} < 0 \Rightarrow$ raises $\Delta G^{\ddagger}$
- **Dissociative:** $\Delta S^{\ddagger} > 0 \Rightarrow$ entropy lowers $\Delta G^{\ddagger}$

Entropy Effects on Barriers – Associative vs. Dissociative

Reaction type	$\Delta n^{\ddagger}$	$\Delta S^{\ddagger}$	**Effect on** $\Delta G^{\ddagger}$	**Example**
Unimolecular	0	≈ 0	Small	Isomerization
Bimolecular (tight TS)	-1	< 0	Raises	S_N2, Diels–Alder
Dissociative	$+1$	> 0	Lowers	Bond homolysis

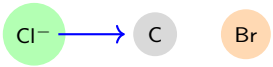
Typical magnitudes:

- Loss of 1 translational DoF: $\Delta S \approx -35$ J/(mol·K) ≈ -8 cal/(mol·K)
- At 298 K: $-T\Delta S \approx +2.4$ kcal/mol per lost translational DoF
- Loss of 1 rotational DoF: $\Delta S \approx -12$ J/(mol·K)

Key insight: For bimolecular reactions, the free energy barrier is **higher** than the electronic barrier due to the entropic penalty of bringing two molecules together.

Case Study 1: S_N2 Reaction

$$Cl^- + CH_3Br \longrightarrow CH_3Cl + Br^-$$



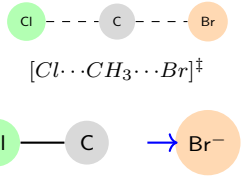
Key features:

- Backside attack by nucleophile
- Walden inversion at carbon
- TS has trigonal bipyramidal geometry
- One-step, concerted mechanism

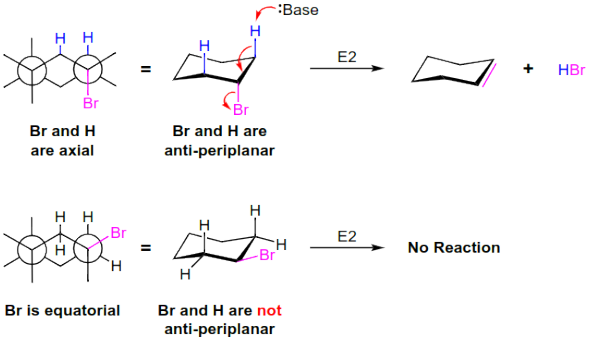
DFT results (B3LYP-D3/def2-TZVP):

- $\Delta E^\ddagger \approx 5$ kcal/mol (gas phase)
- $\Delta G^\ddagger \approx 10$ kcal/mol (solution, $\Delta S^\ddagger < 0$)
- $\Delta G_{\text{rxn}} \approx -8$ kcal/mol (exergonic)

TS:



Case Study 2: E2 Elimination



E2 elimination:

- Concerted: base removes β-H while leaving group departs
- Anti-periplanar geometry required
- One TS, no intermediate

Competition with S_N2:

- Same reactants, different TS
- $\Delta\Delta G^{\ddagger}$ determines selectivity
- Strong base + bulky substrate $\rightarrow$ E2
- Good nucleophile + primary substrate $\rightarrow$ S_N2

Case Study 3: Diels–Alder Cycloaddition

$$\text{Butadiene} + \text{Ethene} \xrightarrow{\Delta} \text{Cyclohexene}$$

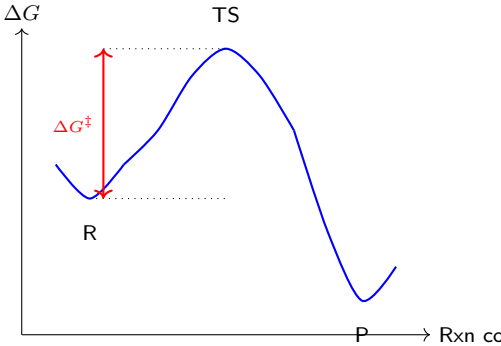
Key features:

4+2 cycloaddition – concerted, pericyclic

- Two new C–C bonds form simultaneously
- TS is **aromatic** (6 electrons in cyclic array)
- Highly ordered TS $\Rightarrow \Delta S^{\ddagger} \ll 0$

DFT results (B3LYP-D3/6-311+G(d,p)):

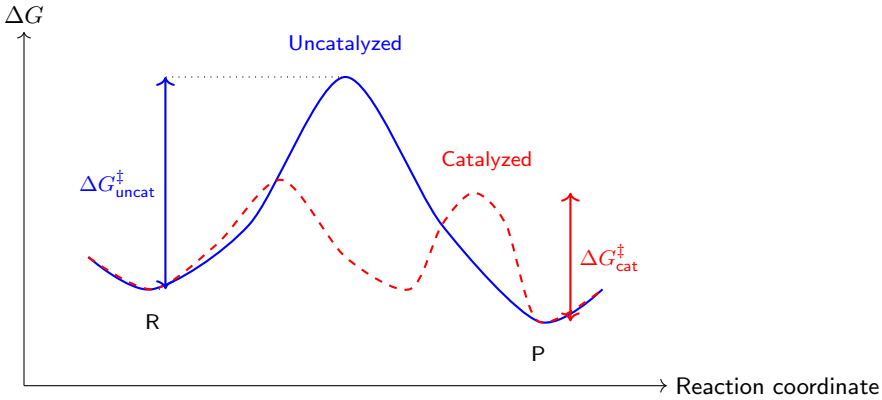
- $\Delta E^{\ddagger} \approx 22$ kcal/mol
- $\Delta H^{\ddagger} \approx 20$ kcal/mol (ZPE correction)
- $\Delta G^{\ddagger} \approx 33$ kcal/mol (large $-T\Delta S^{\ddagger}$!)
- $\Delta G_{\text{rxn}} \approx -38$ kcal/mol



Note: $\Delta S^{\ddagger}$ contributes ~13 kcal/mol to the barrier at 298 K!

Case Study 4: Catalysis – Lowering the Barrier

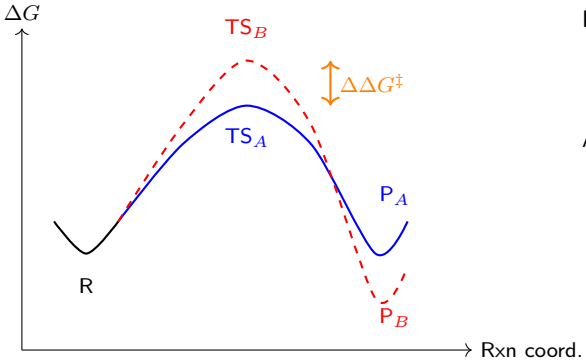
A catalyst provides an alternative pathway with a lower barrier:



- Catalyst lowers $\Delta G^{\ddagger}$ but does **not** change ΔG_{rxn}
- DFT can map the full catalytic cycle
- Compare barriers for catalyzed vs. uncatalyzed paths

Selectivity from Energy Profiles

When multiple products are possible, **selectivity** is determined by $\Delta\Delta G^{\ddagger}$:



Kinetic selectivity:

$$\frac{k_A}{k_B} = \mathrm{e}^{-\Delta\Delta G^{\ddagger}/RT}$$

At 298 K:

$\Delta\Delta G^{\ddagger}$	Selectivity
0.5 kcal/mol	70:30
1.0 kcal/mol	84:16
1.4 kcal/mol	90:10
2.0 kcal/mol	97:3
3.0 kcal/mol	99.4:0.6

Note: Product B is more stable (thermodynamic product) but Product A forms faster (kinetic product). **Kinetic vs. thermodynamic control!**

Thermodynamic vs. Kinetic Control

Kinetic control (low T, short time):

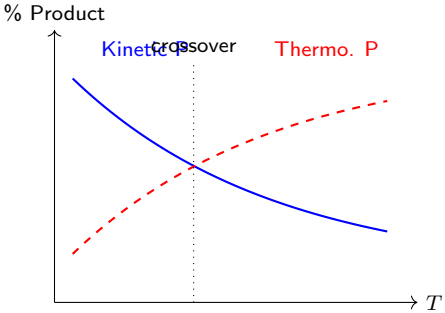
- Product formed **fastest** dominates
- Determined by $\Delta G^{\ddagger}$ (barrier height)
- Irreversible conditions

Thermodynamic control (high T, long time):

- Most **stable** product dominates
- Determined by ΔG_{rxn} (reaction free energy)
- Reversible conditions (equilibrium)

Boltzmann distribution at equilibrium:

$$\frac{N_A}{N_B} = \mathrm{e}^{-(G_A - G_B)/RT}$$



At low T: kinetic product dominates.
At high T: thermodynamic product dominates.

Regioselectivity Example: Electrophilic Aromatic Substitution

Nitration of toluene: Where does NO_2^+ attack?

Position	$\Delta G^\ddagger$ (kcal/mol)	**Relative rate**	**Product %**
ortho	21.5	1.0	∼58%
meta	23.8	0.02	∼4%
para	21.8	0.6	∼37%

DFT can predict:

- Activation barriers for each regioisomeric pathway
- Product ratios from $\Delta\Delta G^\ddagger$
- Effect of substituents on selectivity
- Statistical corrections for equivalent positions (*ortho*: $\times 2$, *para*: $\times 1$)

Stereoselectivity – Enantioselective Catalysis

DFT can predict enantioselectivity by comparing TS energies for R and S products:

$$\text{ee} = \frac{|k_R - k_S|}{k_R + k_S} \times 100\% = \frac{|\mathrm{e}^{-\Delta G_R^{\ddagger}/RT} - \mathrm{e}^{-\Delta G_S^{\ddagger}/RT}|}{\mathrm{e}^{-\Delta G_R^{\ddagger}/RT} + \mathrm{e}^{-\Delta G_S^{\ddagger}/RT}} \times 100\%$$

$\Delta\Delta G^{\ddagger}_{R/S}$ (kcal/mol)	**ee (%)**
0.5	40
1.0	70
1.4	84
2.0	93
3.0	99

Just 2 kcal/mol difference gives >90% ee!
This is within DFT accuracy – predictions are meaningful but require careful methodology.

Choosing the Right Level of Theory for Reactions

Level	Good for	Typical error
B3LYP/6-31G(d)	Quick screening, qualitative trends	3–5 kcal/mol
B3LYP-D3/def2-TZVP	Organic reactions, conformational analysis	2–3 kcal/mol
ωB97X-D/def2-TZVP	General purpose, thermochemistry	1–3 kcal/mol
PBE0-D3/def2-TZVP	Transition metals, organometallics	2–4 kcal/mol
DLPNO-CCSD(T)/cc-pVTZ	"Gold standard" single points	$<$1 kcal/mol

Common strategy: Optimize at DFT, single-point at CCSD(T)

- Geometry: B3LYP-D3/def2-TZVP (cheap, geometries are good)
- Energy: DLPNO-CCSD(T)/cc-pVTZ//B3LYP-D3/def2-TZVP
- Thermal corrections: from the DFT frequency calculation

Solvent Effects

Many reactions occur in solution, not gas phase!

Implicit solvation models (continuum):

- **PCM** (Polarizable Continuum Model) – Gaussian default
- **SMD** (Solvation Model based on Density) – recommended for ΔG_{solv}
- **COSMO/CPCM** – ORCA, Turbomole

Effect on barriers:

- Charged species are strongly stabilized in polar solvents
- S_N1: solvent stabilizes carbocation $\Rightarrow$ **lowers** barrier
- S_N2: solvent stabilizes nucleophile more than TS $\Rightarrow$ **raises** barrier
- Neutral reactions: smaller solvent effects

In Gaussian: `SCRF=(SMD,Solvent=Water)`
In ORCA: `! CPCM(Water)`

Dispersion Corrections – Don't Forget Them!

Standard DFT functionals miss London dispersion forces:

- Attractive van der Waals interactions between all atoms
- Crucial for: conformational energies, reaction barriers involving large molecules, non-covalent interactions

Popular dispersion corrections:

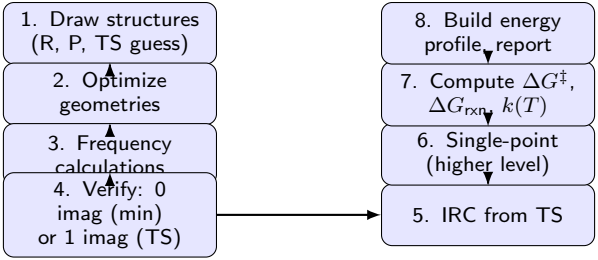
Method	Type	Key features
DFT-D3 (Grimme)	Pairwise additive	BJ or zero damping; most common
DFT-D4 (Grimme)	Charge-dependent	Improved over D3
XDM (Johnson)	Dipole moment model	Non-empirical; my research area!
VV10/NL	Nonlocal functional	Self-consistent; no atom parameters

Rule of thumb: Always use dispersion corrections unless you have a good reason not to.
In Gaussian: `B3LYP/def2-TZVP EmpiricalDispersion=GD3BJ`

Common Pitfalls in Reaction Modeling

1. **Not verifying the TS**
   - Always check: 1 imaginary frequency + IRC
2. **Comparing energies at different levels of theory**
   - All species must use the same method/basis set
3. **Forgetting thermal corrections**
   - $\Delta E \neq \Delta G$! Use ΔG for rate constants
4. **Ignoring conformational sampling**
   - The lowest-energy conformer matters; check multiple conformers
5. **Missing solvent effects**
   - Gas-phase barriers can differ by >10 kcal/mol from solution
6. **Low-frequency mode errors**
   - Modes < 100 cm^{-1} cause large errors in S_{vib} and G
   - Use Grimme's quasi-RRHO or Truhlar's quasi-harmonic correction
7. **Neglecting dispersion**
   - Use D3, D4, or XDM corrections with your functional

Complete Workflow: Studying a Reaction with DFT



Example ORCA Input: TS Optimization

ts_opt.inp

```
! B3LYP D3BJ def2-TZVP OptTS Freq
%maxcore 4000
%pal nprocs 8 end

* xyz 0 1
C   0.000000   0.000000   0.000000
H   0.000000   0.000000   1.089000
H   1.026720   0.000000  -0.363000
H  -0.513360  -0.889165  -0.363000
Cl -0.513360   0.889165  -2.500000
Br  0.513360  -0.889165   2.800000
*
```

Key options:

- `OptTS`: TS optimization using eigenvector following
- `Freq`: frequency calculation to verify the TS
- `D3BJ`: Grimme's D3 dispersion with Becke–Johnson damping

Example Gaussian Input: Complete Reaction Study

Step 1: Optimize Reactant

```
#p B3LYP/def2-TZVP EmpiricalDispersion=GD3BJ Opt Freq SCRF=(SMD,Solvent=Water)
Reactant optimization --- 0 1 --- [coordinates]
```

Step 2: Optimize TS

```
#p B3LYP/def2-TZVP EmpiricalDispersion=GD3BJ Opt=(TS,CalcFC,NoEigenTest) Freq SCRF=(SMD,Solvent=Water)
TS optimization --- 0 1 --- [coordinates from scan or QST]
```

Step 3: IRC

```
#p B3LYP/def2-TZVP EmpiricalDispersion=GD3BJ IRC=(CalcFC,MaxPoints=50,StepSize=10)
IRC from TS --- 0 1 --- [coordinates from optimized TS]
```

Beyond TST: Quantum Tunneling

TST assumes classical nuclear motion – but quantum effects matter!

Quantum tunneling:

- Light atoms (especially H) can **tunnel through** the barrier
- Rate is **faster** than TST prediction
- Important for H-transfer reactions at low temperatures
- Large kinetic isotope effects (KIE): $k_H/k_D \gg 7$

Simple corrections:

- **Wigner correction:** $\kappa_W = 1 + \frac{1}{24}\left(\frac{h|\nu^\ddagger|}{k_B T}\right)^2$
- **Eckart barrier:** fits a 1D potential to reactant, TS, product energies
- For accurate tunneling: need multidimensional approaches (SCT, CVT/SCT)

Example: H-atom transfer in methylhydroxycarbene $\rightarrow$ acetaldehyde occurs entirely by tunneling below 20 K!

Kinetic Isotope Effects (KIE)

Replacing H with D changes the rate – a powerful mechanistic tool:

$$\text{KIE} = \frac{k_H}{k_D}$$

Primary KIE (bond to H/D is broken in the RDS):

- $k_H/k_D \approx 2$–7 (classical, from ΔZPE)
- $k_H/k_D > 7$: suggests tunneling contribution
- DFT prediction: compute $\Delta G^\ddagger$ with H and with D

Secondary KIE (bond to H/D is **not** broken):

- $k_H/k_D \approx 0.8$–1.2
- Reflects changes in hybridization at the TS

DFT can predict KIE: Run frequency calculations with different isotopic masses $\rightarrow$ different ZPE and vibrational partition functions $\rightarrow$ different $\Delta G^\ddagger$.

Conformational Searching and Boltzmann Averaging

Problem: Flexible molecules have many conformers. Which one do you use?

Answer: All of them! Boltzmann-weighted average:

$$\langle A \rangle = \frac{\sum_i A_i \, \mathrm{e}^{-G_i/RT}}{\sum_i \mathrm{e}^{-G_i/RT}}$$

Practical approach:

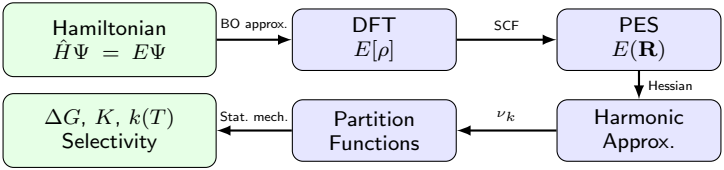
1. Generate conformers (CREST, RDKit, systematic search)
2. Pre-screen at low level (GFN2-xTB, PM7)
3. Optimize unique low-energy conformers at DFT level
4. Compute ΔG for each conformer
5. Boltzmann-weight to get ensemble properties

For TS: You need to search conformational space for the TS too!
Missing a lower-energy TS conformer $\Rightarrow$ overestimate the barrier.

Summary of Lecture 2

1. **Transition states** are first-order saddle points with exactly **one imaginary frequency**
2. **Eyring equation:** $k(T) = \frac{k_B T}{h} \mathrm{e}^{-\Delta G^{\ddagger}/RT}$ connects PES to rate constants
3. **Finding TS:** Relaxed scan $\rightarrow$ QST2/3 or NEB $\rightarrow$ eigenvector following $\rightarrow$ verify!
4. **Always verify:** 1 imaginary frequency + IRC in both directions
5. **Selectivity:** $\Delta\Delta G^{\ddagger}$ of $\sim$2 kcal/mol $\Rightarrow$ $>$95% selectivity
6. **Practical:**
   - Use dispersion corrections (D3/D4/XDM)
   - Include solvent effects (SMD/PCM)
   - Report ΔG, not ΔE – thermal corrections matter!
   - Apply quasi-RRHO for low-frequency modes

The Big Picture: DFT for Reaction Chemistry



From first principles to chemical predictions:
No empirical parameters needed (beyond the choice of functional)!

This is the power of modern computational chemistry.

Key Equations to Remember

Transition State Theory:

$$k(T) = \frac{k_B T}{h} \mathrm{e}^{-\Delta G^{\ddagger}/RT}$$

$$\Delta G^{\ddagger} = G(\mathsf{TS}) - G(\mathsf{R})$$

$$E_a \approx \Delta H^{\ddagger} + RT$$

Selectivity:

$$\frac{k_A}{k_B} = \mathrm{e}^{-\Delta\Delta G^{\ddagger}/RT}$$

TS Characterization:

- $\nabla E = \mathbf{0}$
- Exactly 1 imaginary frequency
- IRC connects R $\leftrightarrow$ P

Free Energy:

$$G = E_{\mathsf{elec}} + E_{\mathsf{ZPE}} + H_{\mathsf{th}} - TS$$

Equilibrium:

$$K = \mathrm{e}^{-\Delta G^{\circ}/RT}$$

Thank you!

Questions?

Alastair J. Price
Guest Lecturer, CHM328H1 S
University of Toronto – Winter 2026

These lectures covered:

1. DFT foundations: Hamiltonian → KS-DFT → PES → Harmonic approx. → Thermodynamics
2. Reactions: Transition states → Eyring equation → Energy profiles → Selectivity

Numerical Example – S_N2 Reaction (Eyring Calculation)

Reaction: $Cl^- + CH_3Br \rightarrow CH_3Cl + Br^-$

DFT Results (B3LYP/6-31+G*):

- Electronic barrier: $\Delta E^\ddagger \approx 5$ kcal/mol
- Free energy barrier: $\Delta G^\ddagger \approx 23$ kcal/mol at 298 K

Eyring Equation: $k(T) = \kappa \cdot \frac{k_B T}{h} \cdot \exp\left(\frac{-\Delta G^\ddagger}{RT}\right)$

At 298 K with $\Delta G^\ddagger = 23$ kcal/mol:

- Pre-exponential: $\frac{k_B T}{h} \approx 6.2 \times 10^{12}$ s^{-1}
- Exponential: $\exp(-23/RT) \approx 1.4 \times 10^{-17}$
- **Result:** $k \approx 8.4 \times 10^{-5}$ s^{-1} (slow in gas phase)

Note: Two molecules combining causes large entropy penalty (~ 18 kcal/mol).

Numerical Example – Diels–Alder (Entropy Effects)

Reaction: $C_4H_6 + C_2H_4 \rightarrow C_6H_{10}$ (cyclohexene)

Experimental barrier: ~ 27 kcal/mol — **DFT (B3LYP):** ~ 24 kcal/mol

Energy Breakdown:

- Electronic: $\Delta E^{\ddagger} \approx 22$ kcal/mol
- ZPE: $+0.5$ kcal/mol
- Thermal: $+0.2$ kcal/mol
- **Entropy:** $\Delta S^{\ddagger} \approx -45$ J/(mol·K) $\Rightarrow T\Delta S^{\ddagger} \approx -13.4$ kcal/mol

Therefore: $\Delta G^{\ddagger} = 22 + 0.2 - (-13.4) \approx 35.6$ kcal/mol

Key insight: Entropy accounts for ~ 13 kcal/mol! Electronic energy alone is not predictive.

Numerical Example – Unimolecular Isomerization

Reaction: HCN $\rightarrow$ HNC (isomerization)

Experimental barrier: 11.6 ± 0.2 kcal/mol — **DFT:** 10.8–12.0 kcal/mol

Energy Breakdown:

- Electronic: $\Delta E^{\ddagger} \approx 10.8$ kcal/mol
- ZPE + Thermal: ~ 0.4 kcal/mol
- Entropy: $\Delta S^{\ddagger} \approx +5$ J/(mol·K) $\Rightarrow$ negligible

Therefore: $\Delta G^{\ddagger} \approx \Delta E^{\ddagger}$ (they're nearly equal!)

Eyring: $k \approx 6.2 \times 10^{12} \times \exp(-10.8/RT) \approx 6.8 \times 10^{4}$ s^{-1}

Comparison: Unimolecular (6.8×10^{4} s^{-1}) vs. bimolecular (8.4×10^{-5} s^{-1}) $\Rightarrow$ entropy penalty is huge!

Entropy Comparison: Unimolecular vs. Bimolecular

Reaction Type	$\Delta S^\ddagger$	**Effect**	**Example**
Unimolecular	≈ 0	Minimal	Isomerization
Bimolecular	-40 to -60 J/(mol·K)	Raises $\Delta G^\ddagger$ by 10–15 kcal/mol	S_N2, D–A
Dissociative	$+40$ to $+60$ J/(mol·K)	Lowers $\Delta G^\ddagger$ by 10–15 kcal/mol	Bond homolysis

At 298 K, $T\Delta S^\ddagger$ ranges from -14 kcal/mol to $+14$ kcal/mol.

Implications:

- S_N2 is slow in gas phase but fast in solution
- Diels–Alder barriers are ~ 10 kcal/mol higher than electronic energy
- Some reactions reverse at high T (entropy-driven)

Moral: ALWAYS compute ΔG, not ΔE.

Method Selection Decision Tree

Quick screening? $\rightarrow$ B3LYP/6-31G(d) or GFN2-xTB (error: 3–5 kcal/mol)

Standard organic chemistry? $\rightarrow$ B3LYP-D3/def2-TZVP or PBE-D3/def2-TZVP (error: 2–3 kcal/mol)

Charge transfer or long-range? $\rightarrow$ ωB97X-D/def2-TZVP or CAM-B3LYP-D3 (error: 1–2 kcal/mol)

Enantioselectivity or high accuracy? $\rightarrow$ DFT geometry + CCSD(T) single-point (error: < 1 kcal/mol)

Large molecules? $\rightarrow$ B3LYP-D3/6-31G(d) geometry, then B3LYP-D3/def2-TZVP single-point

ALWAYS:

- Add dispersion (D3/D4/XDM)
- Include solvent (PCM or SMD)
- Verify TS: exactly one imaginary frequency
- Run IRC in both directions

Pitfall: Missing IRC Verification

What went wrong? Found a structure with exactly one imaginary frequency, declared it a TS. But IRC didn't connect to the expected product.

What happened? The structure was a saddle point, but not the correct TS:

- Maybe it's a TS for a different pathway
- Maybe it's a competing reaction
- Maybe it's a higher-order saddle point with a very small second frequency

The fix:

1. Always run IRC in BOTH directions (forward and backward)
2. Forward direction MUST end at products (same atoms, same connectivity)
3. Backward direction MUST end at reactants
4. If IRC doesn't reach expected structures: the TS is wrong

Why it matters: Without IRC, you don't know what reaction you're studying!

Cost-benefit: IRC takes 5 minutes, saves hours of confusion.

Pitfall: Ignoring Solvent in Ionic Reactions

What went wrong? Computed S_N2 barrier in gas phase: $\Delta G^\ddagger \approx 23$ kcal/mol $\Rightarrow k \approx 10^{-5}$ s^{-1}. Experiment in solution: $\sim 10^3$ s^{-1} (six orders of magnitude faster!).

What happened? In gas phase, the TS is highly polar (two negative charges). DFT doesn't stabilize this. In solution, the solvent dramatically stabilizes the charged TS, lowering $\Delta G^\ddagger$ by 8–10 kcal/mol.

Corrected: $\Delta G^\ddagger(\text{solvent}) \approx 23 - 9 = 14$ kcal/mol

Eyring: $k \approx 10^3$ s^{-1} (matches experiment!)

The fix:

1. For ANY reaction with charged species: ADD SOLVENT
2. Use implicit solvation (PCM or SMD in Gaussian/ORCA)
3. Aqueous: `solvent=water`; Organic: `solvent=acetonitrile`
4. Recompute TS and check $\Delta G^\ddagger$ (often 5–15 kcal/mol lower)

Solvent effects are not small. They're often larger than differences between good and bad methods.